RAJALAKSHMI ENGINEERING COLLEGE

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M2

Started on	Wednesday, 26 August 2026, 1:46 PM
State	Finished
Completed on	Wednesday, 26 August 2026, 2:33 PM
Time taken	46 mins 48 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);
    int count = 0;

    // Count the evaluation of the if condition
    count++;
    if (n == 1) {
        // printf("*");
        count++;
    } else {
        int i = 1;
        while (1) {
            // Count the outer loop condition check (i <= n)
            count++;
            if (!(i <= n)) {
                break;
            }
            // Inner loop: init j=1 (declaration, not counted)
            // Count the inner loop condition check (j <= n) — executes once
            count++;
            // Body executes once due to break
            // printf("*");
            count++;
            // printf("*");
            count++;
```

```
            // break; (not counted)
            // Outer loop update i++
            i++;
            count++;
        }
    }

    printf("%d", count);
    return 0;
}
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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